Bilinear Mixing Model-Based Spectral Decomposition Deep Neural Network for Hyperspectral Target Detection

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***Abstract*— Hyperspectral image (HSI) target detection is an advanced technology within the remote sensing society, widely applied in both civilian and military domains. However, subpixel target detection poses a significantly challenging problem in HSI processing. Existing methods for HSI target detection mostly rely on constructing detectors based on a linear mixing model (LMM). Nevertheless, due to multiple scattering, LMM fails to capture the nonlinear features and extensive interactions among various materials commonly present in real HSIs, potentially resulting in poor performance. To address this issue, this work proposes a novel method for hyperspectral target detection (HTD) based on a bilinear mixing model (BMM) in HSI. Specifically, we designed a spectral decomposition network to separate the background and target. To further extract and combine spectral and spatial information of the image for more accurate learning of target and background distributions, a deep network model based on a multiscale feature extraction module is proposed. Finally, to enhance the discrimination between targets and backgrounds, a customized CEM loss is formulated by minimizing the response of the background and maximizing the response of the target. Experimental comparisons on six real HSI datasets validate the superiority of the proposed detector.**

***Index Terms*— Background learning, bilinear mixing model (BMM), hyperspectral image (HSI), spectral decomposition, target detection.**

I. Introduction

HYPERSPECTRAL imaging refers to a data cube in three dimensions comprising hundreds of spectral bands. Leveraging its rich spectral information, objects of different materials in the image can be identified. To date, hyperspectral imaging has been widely applied in different fields such as classification [1], anomaly detection [2], [3], change detection [4], and target detection [5], [6], [7]. Hyperspectral target detection (HTD) has garnered significant attention due to its crucial role in real-life applications such as forest fire monitoring, soil analysis, mineral exploration, and military deployment, presenting excellent application advantages in both civilian and military domains. HTD is fundamentally a binary classification task, involving the determination of whether the pixel under scrutiny belongs to the target material of interest. Its objective lies in leveraging the discriminative information from the target prior spectrum, precisely localizing targets amidst complex backgrounds by analyzing the distinctive spectral characteristics of different materials present in hyperspectral images (HSIs).

A. Backgrounds

The strong demand for applications and rapid technological advancements have resulted in the rapid development of several HTD techniques. Manolakis et al. [8] constructed a matched filter detector. Adaptive matched filter [9] and adaptive cosine estimator (ACE) [10] are founded on different Gaussian distribution probability models. By decomposing the signal of each pixel, targets can be distinguished from interferences. The orthogonal subspace projection (OSP) algorithm [11] detects targets by suppressing undesired background components in the spectrum and enhancing the desired target components. Because of the good detection performance of constrained energy minimization (CEM) [12], many detectors inspired by CEM have been developed gradually. Zou and Shi [13] proposed the hierarchical CEM (hCEM) detector. Zhao et al. [14] introduced an ensemble-based CEM (E-CEM) method. Ma and Chang [15] proposed a kernel-based CEM method. Recently, Chang [16] found that iterative kernel CEM (IKCEM) has even better performance.

Kernel-based methods enhance target detection by mapping data into a higher dimensional space, thereby leveraging the high-order information within this space to capture nonlinear correlations among spectral bands more effectively than traditional detection methods. Kernel versions of classical algorithms, such as kernel spectral matched filter (KMF) [17] and kernel orthogonal subspace projection (KOSP) [18], have been developed, often yielding superior detection performance due to their enhanced ability to handle nonlinear issues. On the other hand, methods based on sparse representation reconstruct HSI pixels via a linear combination of dictionary elements and

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detect targets based on the reconstructed abundances. Combined sparse and collaborative representation (CSCR) [19] employs sparse representations of target atoms and collaborative representations of background atoms to formulate test pixels. Shen et al. [20] proposed a method constrained by dual sparsity (DSC), which employs clustering to create a background dictionary and combines it with sparse representation for HTD. Although these methods have the potential to enhance detection accuracy, they still face challenges related to problem resolution difficulties and low efficiency. Moreover, these methods tend to struggle to effectively capture deeper features inherent in the redundant high-dimensional HSI data.

The surge in application demands and technological advancements has propelled the swift development of deep learning. This progress has significantly advanced deep learning-based HTD methods. Du and Li [21] utilized convolutional neural networks (CNN) to classify each pixel. Xu et al. [22] use a cognitive fusion of GNN and convolutional neural networks (CNNS) to enhance HTD. Jiao et al. [23] explored a novel triplet spectral-wise transformer-based target detector for HTD tasks and used data enhancement methods to build sufficient and balanced training samples for balanced learning. Wang et al. [24] innovatively combine transformer-based encoders and momentum encoder networks to enhance feature extraction capabilities. Generative adversarial networks (GAN) [25] reconstruct images through adversarial learning between a generator and a discriminator, guiding image reconstruction with the discriminator to achieve better reconstruction quality. Gao et al. [26] generated additional target and background samples using GAN. Compared to traditional reconstruction methods, generative models can capture the essential features of the original HSI and reconstruct them into a feature space more conducive to detection. Kingma and Welling [27] introduced a variational autoencoder (VAE) based on an autoencoder (AE), transforming the latent variable prediction into the latent variable distribution's parameter prediction. Qin et al. [28] propose a two-stage detection framework based on adversarial learning to extract spectral features in latent space through background reconstruction under weak supervision. Some approaches combine classical reconstruction methods with generative models for improved reconstruction effects. Shen et al. [30] decomposed the spectrum based on the linear mixing model (LMM) [29] and optimized the network with a CEM loss for target detection. To address the issues of spectral variation and insufficient model generalization, Shen et al. [31] proposed the HTD-Mamba method, which for the first time applies Mamba to HTD. Luo et al. [32] further enhanced the model's generalization ability through the SelfMTL method, which employs self-supervised metatransfer learning and contrastive representation.

B. Motivations

Despite the significant achievements of deep learning-based methods, several challenges persist. For example, many deep learning detection methods [21], [33] focus on extracting features from individual spectral information, often neglecting the spatial context of images. Moreover, direct detection methods reliant on deep learning typically demand a large volume of high-quality labeled samples for training. Acquiring a substantial amount of high-quality labeled spectral samples in specific scenarios remains a formidable challenge in practical applications. Limited consideration of prior spectral information could result in trained models having inadequate generalization capabilities when detecting real HSI data sampled under varying scenes. While existing efforts utilizing generative models like AE [33], GAN [26], or the LMM [34] for sample generation have mitigated some of these issues to an extent, these sample generation methods often struggle to simulate the nonlinear interactions among multiple materials caused by multiple scattering [35], which are prevalent in the hyperspectral imaging process. Detection methods based on background learning [36], employing generative models to reconstruct the background with target suppression constraints applied during the reconstruction process, have somewhat addressed the challenge of simulating target samples. Nevertheless, these techniques primarily rely on the detection accuracy of traditional techniques and struggle to precisely separate the background from targets.

C. Contributions

To overcome these challenges, we introduce a novel subpixel target detection model based on the bilinear mixing model (BMM) for spectral decomposition. Due to nonlinear scattering effects during the image acquisition process, separating complete targets from images presents a significant challenge. Existing HTD methods are mostly based on LMM. However, the LMM alone is insufficient for isolating pure targets because it does not account for these nonlinear scattering phenomena, leading to a decrease in model accuracy when dealing with multiple scattering effects in HSIs. To address this challenge, we employ BMM for more effective separation. The BMM can accurately represent nonlinear scattering by simulating virtual endmembers and their abundances to represent the noise components generated by nonlinear scattering, therefore capturing the interactions [37], [38] among numerous materials. By isolating and removing these noise components, the separated targets can be brought closer to their true form. This approach ensures that detected targets are as close as possible to their actual morphology, thereby enhancing detection accuracy and reliability. While the linear branch captures the primary target information, the BMM allows us to mitigate the impact of nonlinear scattering, leading to a more accurate representation of the targets. This approach enables the model to better learn the contributions of different endmembers and their interrelationships. Compared with the LMM, BMM enables more accurate fitting of spectra with nonlinear effects and improves target detection accuracy. As revealed by Somers et al. [39], interactions of third-order and higher orders can be neglected. Consequently, the advantages of BMM lie in its focus on critical nonlinear effects, balance between model complexity and authenticity, efficient computation, and strong physical interpretability, making it more suitable for applications involving significant second-order interactions in ground object mixtures. Existing

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BMM-based unmixing methods primarily extract endmembers using methods like vertex component analysis (VCA) or directly obtain them from laboratory data. Endmember-guided unmixing network (EGU-Net) [40] employs VCA to extract endmembers from HSIs. Multiobjective endmember extraction method with the bilinear mixture model (Bi-MoEE) [41] selects real and virtual endmembers from an extended spectral library composed of standard spectral libraries and their virtual products. These endmember extraction methods can lead to complex models and insufficient generalization capabilities. In contrast, our approach initializes the endmember matrix randomly and adjusts the endmember information during the training process. In addition, we introduce a reconstruction loss to guide the optimization direction of the endmember information, ensuring the reliability and robustness of the method. The primary contributions of our study are threefold.

1) We propose a novel spectral decomposition network based on BMM that combines deep learning with traditional theory. Compared with traditional BMM-based spectral decomposition networks, we fully leverage the advantages of deep learning by randomly initializing the background endmember matrix and gradually optimizing it during the network training process. This approach effectively removes nonlinear scattering interference, yielding purer target components.
2) To fully exploit both spectral and spatial information within the images, we introduce a multiscale feature extraction module called U-net and pyramid squeeze attention (U-PSA). This module enhances the capability of the spectral decomposition network to extract nonlinear features from HSIs. Experimental results demonstrate that U-PSA outperforms existing advanced feature extraction modules, achieving superior performance.
3) For more precise separation of targets and backgrounds, we incorporate CEM loss to optimize the model. In addition, during training, we introduce sparsity loss and reconstruction loss to ensure the physical interpretability of BMM and the reliability and robustness of the model. Jointly optimizing with various loss functions, our method successfully achieves precise separation between background and target components.

D. Organizations

The remainder of the article is organized as follows. Section II presents the preliminaries of the BMM and CEM detector. Section III provides a detailed description of the proposed novel HSI target detection method. In Section IV, we report and analyze the experimental results obtained on six real-world HSI datasets. Finally, the conclusion and future work are summarized in Section V.

II. Notations and Preliminaries

A. Notations

Hereinafter, scalars, vectors, matrices, and tensors are expressed by x, $\mathbf{x}$, $\mathbf{X}$, and $\mathcal{X}$, respectively. For example, the third-order HSI tensor is signified by $\mathcal{X} \in \mathbb{R}^{h\times w\times l}$, where h, w, and l are the height, width, and spectral bands, respectively. The unfolded spectral matrix of $\mathcal{X}$ and the prior target spectrum are signified by $\mathbf{X} \in \mathbb{R}^{l\times n}$ and $\mathbf{d} \in \mathbb{R}^{l\times 1}$, where n is the pixels number of HSI, i.e., $n = h \times w$.

B. Bilinear Mixing Model

The BMM accounts for the nonlinear scattering effects that occur during imaging. It can accurately represent nonlinear scattering by simulating virtual endmembers and their abundances, therefore capturing the interactions among numerous materials. Specifically, it introduces virtual endmembers to represent the noise components generated by nonlinear scattering within the image. By isolating and removing these noise components, the separated targets can be brought closer to their true form. This approach ensures that detected targets are as close as possible to their actual morphology, thereby enhancing detection accuracy and reliability. BMM treats second-order photon interactions between different endmembers as additional terms in the LMM, assuming that third-order or higher order interactions are negligible [42]. According to the principle of BMM, a spectral matrix with l bands can be expressed as follows:

$$\mathbf{X} = \mathbf{AS} + \mathbf{HV} + \mathbf{I}, \ \text{s.t. } \mathbf{S}, \mathbf{V} \geq 0, \ \mathbf{E}_{m+1}^{\mathrm{T}}\mathbf{S} = \mathbf{E}_n^{\mathrm{T}} \tag{1}$$

where $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_{m+1}] \in \mathbb{R}^{l\times(m+1)}$ is the endmember spectral matrix. Assuming, without losing generality, that the first endmember of $\mathbf{A}$ is the target prior $\mathbf{d}$ and the rest of the endmember form the background endmember matrix $\mathbf{A}_b = [\mathbf{a}_2, \mathbf{a}_3, \ldots, \mathbf{a}_{(m+1)}] \in \mathbb{R}^{l\times m}$. Thus, the endmember matrix can also be expressed as $\mathbf{A} = [\mathbf{d}, \mathbf{A}_b]$. $\mathbf{S} = [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_{(m+1)}]^{\mathrm{T}} \in \mathbb{R}^{(m+1)\times n}$ is the abundance matrix, and each column represents the abundance of each pixel, where $\mathbf{s}_1$ denotes the target abundance matrix. Thus, the background abundance matrix can be expressed as $\mathbf{S}_b = [\mathbf{s}_2, \mathbf{s}_3, \ldots, \mathbf{s}_{(m+1)}] \in \mathbb{R}^{m\times n}$.

$\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_p] \in \mathbb{R}^{l\times p}$ denotes the virtual endmember matrix that contains p virtual endmembers. $\mathbf{V} \in \mathbb{R}^{p\times n}$ is the interaction abundance matrix, representing the interaction of multiple materials with nonlinear scattering. $\mathbf{I} \in \mathbb{R}^{l\times n}$ is the error matrix which represents model error, measurement error, and noise. The constraints illustrate abundance sum-to-one constraint (ASC) and the abundance nonnegativity constraint (ANC), which are widely used in physical interpretation, where $\mathbf{E}_{(m+1)} = [1, 1, \ldots, 1]^{\mathrm{T}} \in \mathbb{R}^{(m+1)}$, $\mathbf{E}_n = [1, 1, \ldots, 1]^{\mathrm{T}} \in \mathbb{R}^n$. $[\cdot]^{\mathrm{T}}$ represents the transpose of a vector or matrix. The virtual endmembers matrix $\mathbf{H}$ can be computed from the m endmembers in the endmembers matrix $\mathbf{A}$ by

$$\mathbf{h}_{i\times j} = \mathbf{a}_i \otimes \mathbf{a}_j = \begin{pmatrix} \mathbf{a}_{1,i} \\ \vdots \\ \mathbf{a}_{l,i} \end{pmatrix} \otimes \begin{pmatrix} \mathbf{a}_{1,j} \\ \vdots \\ \mathbf{a}_{l,j} \end{pmatrix} = \begin{pmatrix} \mathbf{a}_{1,i}\mathbf{a}_{1,j} \\ \vdots \\ \mathbf{a}_{l,i}\mathbf{a}_{l,j} \end{pmatrix} \tag{2}$$

where $\otimes$ denotes the Hadamard product, i.e., element-wise multiplication. $\mathbf{a}_i$ and $\mathbf{a}_j$ denote the ith and the jth endmembers of the endmember matrix, respectively, and $i \neq j$. Therefore, the value of the virtual endmember p can be calculated by $p = m(m+1)/2$.

In the proposed model, endmembers are defined as trainable parameters, initialized with random values through a

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random initialization strategy, and dynamically adjusted during the training process alongside the overall optimization of the model. This adaptive optimization mechanism drives the endmember parameters to continuously approach the actual physical endmembers, thereby enabling the second-order interaction relationships characterized by the derived virtual endmembers to continuously converge during the model's training iterations and gradually achieve precise matching with the material mixing characteristics in real-world applications.

Compared to the LMM, BMM considers the second-order scattering of photons between two materials [43]. This model can describe the nonlinear interaction between different materials and matches well with real HSIs. The target detection issue can be reformed as an optimization model for estimating abundance coefficients based on (1). This model can be stated as

$$\begin{aligned}\min_{\mathbf{S}_b,\mathbf{s}_1,\mathbf{H},\mathbf{V}} &\frac{1}{2}\|\mathbf{X}-\mathbf{A}_b\mathbf{S}_b-\mathbf{d}\mathbf{s}_1-\mathbf{H}\mathbf{V}\|_F^2\\ &+\lambda_1\sigma_1(\mathbf{S}_b)+\lambda_2\sigma_2(\mathbf{s}_1)\\ \text{s.t.}\quad &\mathbf{S}_b,\mathbf{s}_1,V\geq 0,\mathbf{E}_m^{\mathrm{T}}\mathbf{S}=\mathbf{E}_n^{\mathrm{T}}.\end{aligned}\tag{3}$$

The first term represents the fidelity term, ensuring the authenticity of the BMM model. The other two terms, $\sigma_1(\mathbf{S}_b)$and $\sigma_2(\mathbf{s}_1)$, respectively, are prior terms used to constrain the background and target coefficients, where λ_1 and λ_2 are weighting parameters. Various prior constraints can be employed to characterize the background and target coefficients. For background abundance, a low-rank prior can reduce linear correlations among background components, a total variational prior ensures local smoothness of the background, and a sparse prior guarantees that each pixel is composed of a small number of endmembers. Similarly, for target abundance, sparse priors ensure the sparsity of targets in the image. These priors can be used separately or in a flexible way to combine to characterize particular features of the target and background. In addition, in a physical sense, endmember abundance and virtual endmember abundance must be nonnegative, and abundance coefficients must comply with the constraint of summing to one.

C. Constrained Energy Minimization

CEM aims to design a finite impulse response filter $\mathbf{w}=[w_1,w_2,\ldots,w_l]^{\mathrm{T}}$ such that the background output energy is minimized while the target output energy is maximized. Therefore, it can be formulated as the following optimization problem:

$$\min_{\mathbf{w}}\mathbf{w}^{\mathrm{T}}\mathbf{X}.\tag{4}$$

Then, its average output energy on the entire image scene is expressed as

$$\min_{\mathbf{w}^{\mathrm{T}}}\frac{1}{n}\left(\mathbf{w}^{\mathrm{T}}\mathbf{X}\right)^2=\min_{\mathbf{w}^{\mathrm{T}}}\frac{1}{n}\mathbf{w}^{\mathrm{T}}\mathbf{X}\mathbf{X}^{\mathrm{T}}\mathbf{w}^{\mathrm{T}}.\tag{5}$$

Now, we can formulate the following constrained optimization problem as follows:

$$\begin{aligned}\min_{\mathbf{w}}&\frac{1}{n}\mathbf{w}^{\mathrm{T}}\mathbf{X}\mathbf{X}^{\mathrm{T}}\mathbf{w}\\ \text{s.t.}\quad&\mathbf{w}^{\mathrm{T}}\mathbf{d}=1.\end{aligned}\tag{6}$$

According to the method of Lagrange multipliers, this problem can be rephrased as

$$\min_{\mathbf{w}}\frac{1}{n}\mathbf{w}^{\mathrm{T}}\mathbf{X}\mathbf{X}^{\mathrm{T}}\mathbf{w}+\mu\left(\mathbf{w}^{\mathrm{T}}\mathbf{d}-1\right)\tag{7}$$

where μ represents the Lagrange multiplier. By setting the derivatives of μ and $\mathbf{w}$ to zero, the optimal solution $\mathbf{w}^*$ can be obtained as follows:

$$\mathbf{w}^*=\frac{\mathbf{d}^{\mathrm{T}}\mathbf{R}^{-1}}{\mathbf{d}\mathbf{R}^{-1}\mathbf{d}^{\mathrm{T}}}\tag{8}$$

where $\mathbf{R}=(1/n)\mathbf{X}\mathbf{X}^{\mathrm{T}}$ is the autocorrelation matrix, and the detection result of CEM is denoted by $\mathbf{w}^{\mathrm{T}}\mathbf{X}$. Building upon the concept of CEM, this article devises a CEM loss to maximize target output and minimize background output. The detailed approach will be elaborated in Section III.

III. Proposed Method

In this section, we first provide a detailed explanation of the proposed approach. Then, we describe in detail the two stages of the detection model, including the spectral decomposition stage and the target detection stage. Finally, we introduce the spectral decomposition network model and the loss functions.

A. Overall Architecture

In this article, we introduce a novel target detection framework based on spectral decomposition and multiscale feature extraction. Leveraging the BMM theory, we successfully decompose the image into subspaces corresponding to the target and background. To fully exploit both spectral and spatial information within the images, we integrate two network architectures: U-PSA module [44]. Ultimately, by jointly optimizing multiple loss functions, we achieve a well-trained detection model. Fig. 1 provides a schematic of the proposed detection framework. The entire framework takes 3-D HSI as input and outputs a target detection map. The framework primarily consists of two components: the BMM-based spectral decomposition component and the target detection component.

The spectral decomposition component of the network can be divided into an encoder and a decoder. Initially, the input HSI undergoes processing through convolutional layers to extract low-level features. Subsequently, the HSI is fed into two independent spectral decomposition networks, U-PSA, forming the encoder to extract and enhance features. The enhanced data are then mapped into endmember abundance and virtual endmember abundance coefficient spaces. The endmember matrix, composed of the target prior and background endmembers, is obtained using (2) to derive the virtual endmember matrix. The members of the endmember matrix are trained alongside the network. Given the nature of network training, we do not impose special initial settings on the end members. Since these are treated as trainable parameters, their initial values do not influence the final detection results and are therefore initialized using random functions. To ensure nonnegativity constraints, rectified linear unit (ReLU) activation functions are employed to obtain the final endmembers. Finally, the decoder performs matrix multiplication of the two

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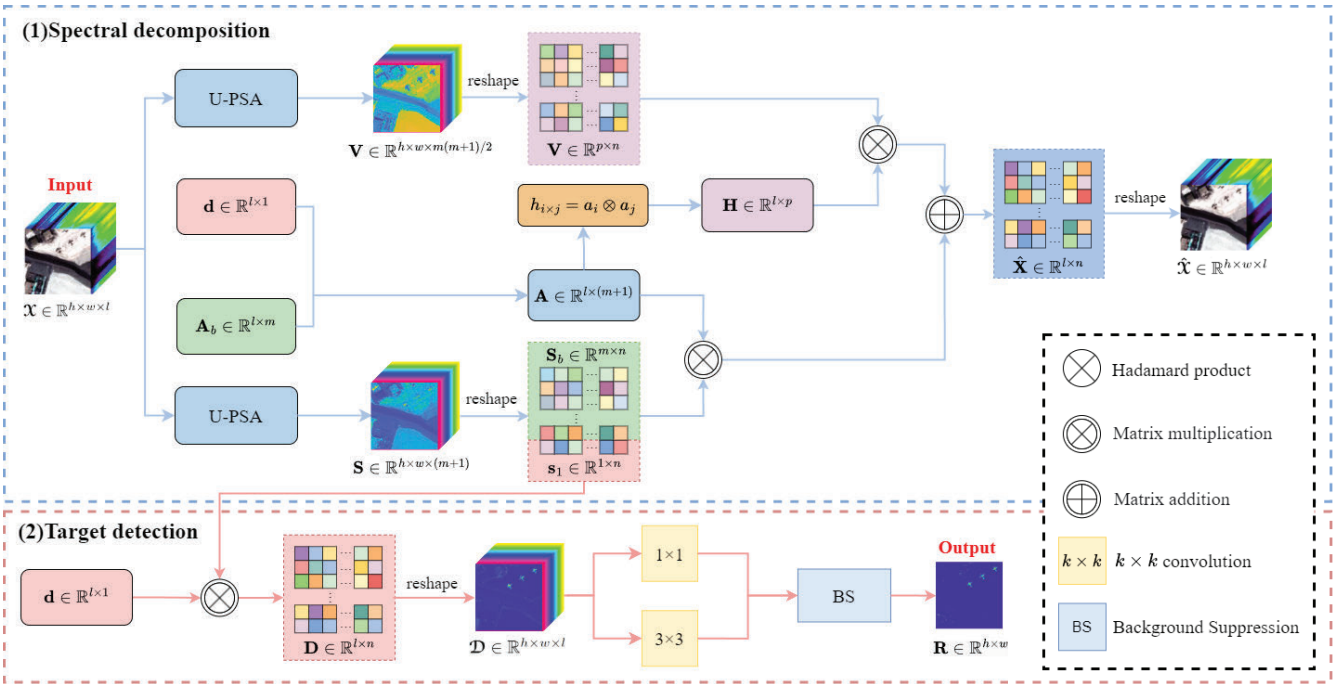


Fig. 1. Schematic of the proposed HSI target detection method.

sets of endmembers and their corresponding abundances based on BMM. The two results are added together to produce the reconstructed HSI. It is important to note that the reconstructed HSI serves as a validation of the spectral decomposition's fidelity rather than being the output of our detection model. A detailed explanation of this process will be provided in Section V.

The target detection component operates in parallel with the spectral decomposition component. After obtaining the target abundance from the previous stage, it is multiplied by the target spectral vector to synthesize the target component for each pixel in the HSI. To fully utilize the spatial information from neighboring pixels, the target components pass through multiscale convolutional layers. Following the multiscale convolutional layers, we introduce a novel background suppression module (BSM). The BSM dynamically recalibrates the features by generating a spatial attention mask. This mask amplifies target-related regions while suppressing background interference through trainable scaling and biasing operations. The resultant refined features are then used to generate the final detection map, ensuring robust performance in cluttered environments.

B. Spectral Decomposition

Our detection framework combines the BMM theory with deep neural networks through a spectral decomposition network and utilizes loss functions to constrain fidelity and prior terms, thus transforming model (3) into an encoder–decoder network structure. This detection model can effectively simulate the widely existing nonlinear scattering in HSI while ensuring both authenticity and physical interpretability. Once the model is determined, optimization of the target detection model can be easily and effectively achieved through gradient descent.

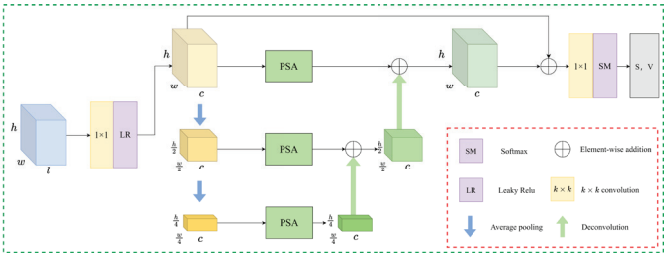


Fig. 2. Schematic of the proposed multiscale feature extraction module U-PSA.

1) Extracting Abundance Coefficients: The spectral decomposition network is designed to learn the nonlinear mapping between the image and abundance coefficient, represented as

$$\begin{aligned}\mathbf{S} &= \mathcal{F}_{\text{sd}}(\mathcal{X}, \theta_1)\\ \mathbf{V} &= \mathcal{F}_{\text{sd}}(\mathcal{X}, \theta_2)\end{aligned} \tag{9}$$

where $\mathcal{F}_{\text{sd}}$ represents the encoder to estimate the abundance from the image, θ_1, and θ_2 denote the parameters involved in the two encoders, respectively.

As shown in Fig. 2, the spectral decomposition network adopts a U-shaped design and utilizes a PSA module for multiscale feature extraction. Specifically, first, a 1×1 convolution is applied to the input HSI for initial feature extraction, mapping the image dimension from l to c dimensions. Then, the leaky ReLU activation function [45] is employed to enhance the nonlinear features. Therefore, the initial features can be represented as

$$\begin{aligned}\mathcal{X}_1 &= \mathcal{LR}\left(\omega_1^1 \mathcal{X}\right)\\ \mathcal{X}_1{}' &= \mathcal{LR}\left(\omega_1^2 \mathcal{X}\right)\end{aligned} \tag{10}$$

where $\mathcal{LR}(\cdot)$ represents the leaky ReLU activation function, ω_i^j represents jth $i \times i$ convolution operation. Following this, we designed a U-shaped structure of the multiscale PSA

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module to further enhance and extract features, which can be represented as

$$\begin{aligned} \mathcal{X}_2 &= \mathrm{UPSA}(\mathcal{X}_1) \\ \mathcal{X}_2{}' &= \mathrm{UPSA}\big(\mathcal{X}_1{}'\big) \end{aligned} \tag{11}$$

where $\mathcal{X}_2$ and $\mathcal{X}_2'$ represent the two sets of enhanced features obtained after being inputted into the UPSA network. Then, the enhanced features are projected into the abundance coefficient space, and the softmax function is used to maintain the ANC and ASC constraints. The two sets of abundance coefficients obtained can be represented as

$$\begin{aligned} \mathbf{S} &= \mathrm{reshape}\big(\mathrm{softmax}\big(\omega_1^3 \mathcal{X}_2\big)\big) \\ \mathbf{V} &= \mathrm{reshape}\big(\mathrm{softmax}\big(\omega_1^4 \mathcal{X}_2{}'\big)\big) \end{aligned} \tag{12}$$

where reshape$(\cdot)$ denotes the operation of transforming the tensor into its corresponding matrix form.

2) Image Restoration: After obtaining $\mathbf{S}$ and $\mathbf{V}$, according to the BMM, the recovered image can be represented as

$$\hat{\mathcal{X}} = \mathrm{reshape}(\mathbf{AS} + \mathbf{HV}) \tag{13}$$

where $\mathbf{A}$ is the endmember matrix, which can be further decomposed into target endmembers $\mathbf{d}$ and background endmembers $\mathbf{A}_b$. Therefore, (13) can be further represented as

$$\hat{\mathcal{X}} = \mathrm{reshape}(\mathcal{RL}(\mathbf{A}_b\mathbf{S}_b) + \mathbf{ds}_1 + \mathbf{HV}) \tag{14}$$

where the background endmembers $\mathbf{A}_b$ are trainable parameters, optimized jointly with the network parameters. Due to the nature of network training, we do not impose specific initial values on the endmembers. Since they are treated as training parameters, their initial values will not affect the final detection results, which are thus initialized by a random function. $\mathcal{RL}(\cdot)$ is the rectified linear unit activation function, used to maintain the nonnegativity property of the background endmembers, to conform to reality. The reshape$(\cdot)$ operation here transforms the matrix form of the recovered image into its corresponding tensor form.

C. Target Detection

After obtaining the target endmember abundance coefficients matrix $\mathbf{s}_1$, the target component $\mathbf{D}$ can be obtained by multiplying the target abundance matrix with the target prior. Further transformation yields its tensor form $\mathcal{D}$, which can be expressed by

$$\mathcal{D} = \mathrm{reshape}(\mathbf{ds}_1). \tag{15}$$

$\mathcal{D}$ effectively represents the pure target spectral components within the image. For why the reshape operation is necessary, we will explain it below. Our spectral unmixing network leverages the mixed characteristics of hyperspectral pixels, which implies that the synthesized target components derived from the output target abundances may contain subpixel targets. Therefore, to transform the target components into the final detection map, it is necessary to establish a mapping from the target components to the detection map. This process requires effectively utilizing spatial correlations within each pixel's neighborhood to more accurately convert the target components into the final detection map. In this context, our approach considers the spatial relationships within the image. Specifically, pixels adjacent to target pixels are more likely to be target pixels themselves, and similarly, pixels adjacent to background pixels are more likely to be background pixels. By constructing dynamic maps using multiscale convolutional layers, we can effectively exploit local spatial correlations. To enable the multiscale convolutional layers to fully utilize the spatial information from each pixel's neighborhood, we perform a reshape operation that converts the target component matrix $\mathbf{D}$ into a 3-D target component tensor $\mathcal{D}$. This transformation ensures that the spatial structure of the data is preserved, allowing the multiscale convolutional layers to operate efficiently and produce a more precise and reliable detection result. The reshape operation facilitates the conversion of the target components into a format that aligns with the spatial dimensions of the original dataset, thereby enhancing the accuracy of the detection map. The preliminary detection result can be represented as

$$\mathbf{D}' = \omega_1^5 \mathcal{D} + \omega_3^1 \mathcal{D} \tag{16}$$

where the preliminary detection result $\mathbf{D}'$ is obtained by combining the outputs of the 1×1 convolutional block and 3×3 convolutional block. This strategy effectively utilizes both block-level spatial information and pixel-level spectral information to detect the targets. To further enhance the background suppression effect, the preliminary detection result $\mathbf{D}'$ is fed into the BSM to obtain the final detection map $\mathbf{R}$, which can be represented as

$$\mathbf{R} = \mathrm{BSM}\big(\mathbf{D}'\big). \tag{17}$$

D. Background Suppression Module

To suppress background interference in hyperspectral target detection, we propose a BSM that adaptively refines preliminary detection maps. Given an initial detection map $\mathbf{D}' \in \mathbb{R}^{H \times W}$, the BSM employs two trainable parameters, a scaling factor β_1 and a bias term β_2, to dynamically generate a spatial weight map. The weight map $\mathbf{W_{BSM}}$ is computed through a nonlinear transformation, which can be represented as

$$\mathbf{W_{BSM}} = \sigma\left(\beta_1 \cdot \mathbf{D}' + \beta_2\right) \tag{18}$$

where $\sigma(\cdot)$ denotes the Sigmoid function, compressing values into the range [0, 1]. This weight map highlights the target regions and suppresses the background noise. The refined detection map $\mathbf{R}$ is then obtained by element-wise multiplication, which can be represented as

$$\mathbf{R} = \mathbf{W_{BSM}} \otimes \mathbf{D}'. \tag{19}$$

E. Multiscale Feature Extraction Network

Inspired by PSA modules and the U-net, we propose a multiscale feature extraction network. It can enhance the feature of the input image and extract the effective information of the image sufficiently. This enables obtaining more useful features for accurately obtaining the abundance coefficients.

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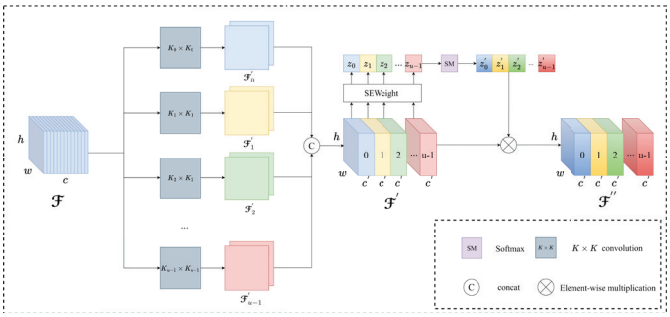


Fig. 3. Schematic of the PSA module.

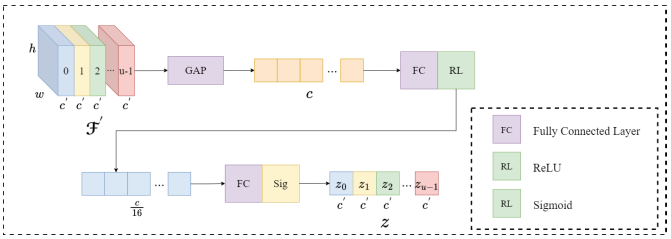


Fig. 4. Schematic of the squeeze-and-excitation (SE) weight module.

1) Overall Structure: As shown in Fig. 2, this U-shaped network structure consists of three layers of feature encoding, and each layer extracts data features at different scales through a PSA module. From the first layer to the last layer, the picture feature's resolution is decreased by using the average pooling operations, followed by the transpose convolution operations to restore the low-resolution features to the high-resolution space. This multiscale approach captures contextual information at different resolutions, enhancing the representation of both local and global features. Within each scale, spatial and channel attention mechanisms are applied to emphasize important regions and channels. Spatial attention focuses on significant spatial locations, while channel attention highlights relevant spectral bands. This dual attention mechanism ensures that the most informative features are retained and enhanced. To minimize data loss throughout the restoration procedure, this article integrates the output of the encoded feature by the PSA with the low-resolution bottom-up reconstruction features. By integrating enhanced features obtained at multiple scales, the primary information of the HSI can be further retained. In addition, to reduce computational complexity, the number of channels in the image remains unchanged during average pooling and transpose convolution operations.

2) Pyramid Squeeze Attention Module: The PSA module employs a pyramid structure that processes the HSI at multiple scales. It integrates spatial and channel attention mechanisms to extract robust features from both spectral and spatial dimensions. This module is highly advantageous for fully extracting the spectral and spatial features of the image. As shown in Fig. 3, the PSA module mainly consists of four steps. First, spatial information of the input feature maps $\mathcal{F}$ is extracted in a multibranch fashion, where the input channel dimension of each branch is c. This approach enables the acquisition of richer multiscale spatial information from the image. After convolution with different scale kernels, feature maps at various scales are obtained, which can be represented as

$$\mathcal{F}_i' = \omega_{k_i}^1 \mathcal{F}, \quad i = 0, 1, 2, \ldots, u-1 \tag{20}$$

where $\mathcal{F}_i'$ represents the ith multiscale feature map. They all have the same channel dimension $c' = c/u$. Each branch can locally establish cross-channel interactions and autonomously learn multiscale spatial information. The entire multiscale feature map $\mathcal{F}'$ is obtained by concatenating different multiscale feature maps $\mathcal{F}_i'$.

Then, different-scale attention weight vectors are produced by extracting channel attention weight information from the multiscale preprocessed feature map. Specifically, the SEweight module is used to extract attention from multiscale feature maps, obtaining channel-level attention vectors, denoted as

$$\mathbf{z}_i = \text{SEweight}(\mathcal{F}_i'), \quad i = 0, 1, 2, \ldots, u-1 \tag{21}$$

where SEweight$(\cdot)$ is designed to encode global information and adaptively recalibrate channel weights. Through this mechanism, the model can be trained to selectively emphasize beneficial spectral features and reduce interfering noise features. Specifically, as shown in Fig. 4, to more efficiently integrate linear information between channels, channel statistics are created using global average pooling and then transferred through two fully connected layers. Finally, the sigmoid activation function is applied to obtain channel weight information. Then, the channel-level attention vectors are recalibrated using the softmax function, which can be expressed as

$$\mathbf{z}_i' = \text{softmax}(\mathbf{z}_i) = \frac{\exp(\mathbf{z}_i)}{\sum_{i=0}^{u-1} \exp(\mathbf{z}_i)}, \quad i = 0, 1, 2, \ldots, u-1. \tag{22}$$

This approach allows the PSA module to fully utilize the spectral and spatial information of the image by integrating features from various scales. Ultimately, the output is a refined feature map with richer multiscale feature information, which is created by multiplying the recalibrated weights with the appropriate feature maps. This is indicated as follows:

$$\mathcal{F}_i'' = \mathcal{F}_i' \mathbf{z}_i', \quad i = 0, 1, 2, \ldots, u-1. \tag{23}$$

The output result $\mathcal{F}''$ is obtained by concatenating each $\mathcal{F}_i''$ together.

F. Loss Function

Our spectral decomposition deep network is built based on the BMM theory. Therefore, we first design a reconstruction loss to ensure the authenticity of spectral decomposition. Specifically, the image reconstruction loss measures the difference between the reconstructed HSI and the original HSI to ensure that the endmembers and abundances generated by the decomposition network satisfy the distribution of real images. We choose the $L1$-norm to quantify the difference between the input HSI and the reconstructed HSI since it can better maintain the edge and texture information of the

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image. By following this strategy, the reconstruction loss can be represented as follows:

$$\mathcal{L}_{\text{recon}} = \frac{1}{\text{hwl}} \sum_{i=1}^{n} \left| \mathbf{x}_i - \hat{\mathbf{x}}_i \right| \tag{24}$$

where $\mathbf{x}_i$ and $\hat{\mathbf{x}}_i$ represent the ith pixel of the input HSI and the ith pixel of the reconstructed HSI, respectively. $|\cdot|$ represents the absolute value.

Furthermore, motivated by the central concept of CEM, which aims to significantly improve detection accuracy by minimizing the output energy of background and maximizing the output energy of known target spectra, this article specifically designs a CEM loss to suppress background and highlight the target. It is well-known that a target detector can be viewed as a mapping from l-dimensional space to 1-D space. Therefore, the deep network detector proposed in this article can be regarded as a response filter in CEM. Thus, the CEM loss can be stated as

$$\mathcal{L}_{\text{CEM}} = \frac{\alpha_1}{\text{hw}} \|\mathcal{F}(\mathcal{X})\|_F^2 + |\mathcal{F}(\mathbf{d}) - 1| \tag{25}$$

where α_1 is the balance parameter, $\mathcal{F}(\mathcal{X})$ and $\mathcal{F}(\mathbf{d})$, respectively, represent the detection result of the HSI and the target. $||\cdot||_F$ represent the Frobenius norm. In the entire equation, the first term relates to the CEM optimization term, which minimizes the output of the background, and the second term relates to the CEM constraint, which is to force the target spectrum's output to equals 1 to maximize the output of the target. Since the target prior often consists of only one spectrum, for the convenience of calculation, we specifically design a tensor cube $\mathcal{T} \in \mathbb{R}^{h \times w \times l}$ consisting entirely of target vectors, which has the same dimensions as the test image, to maximize its output through the deep detector. Following this strategy, the detection map and the detection map of the target can be defined as follows:

$$\mathcal{R} = \mathcal{F}(\mathcal{X}),\ \mathcal{R}_d = \mathcal{F}(\mathcal{T}). \tag{26}$$

However, the energy minimization constraint treats each pixel equally, without specific discrimination between background and target, which may result in poor detection performance. To address this challenge, we use a weight matrix to represent the differences between the target prior and each pixel. Hence, the CEM loss can be expressed as

$$\mathcal{L}_{\text{CEM}} = \frac{1}{\text{hw}} \sum_{i=1}^{h} \sum_{j=1}^{w} \left(\alpha_1 \mathbf{O}(i, j) \mathcal{R}^2(i, j) + |\mathcal{R}_d(i, j) - 1| \right) \tag{27}$$

where $\mathcal{R}(i, j)$ corresponds to the output energy from the CEM process, which we aim to minimize. The primary objective of the CEM loss function is to minimize the output energy associated with background components. This ensures that nontarget elements are suppressed, enhancing the visibility of target signatures. In addition, the loss function includes a constraint that forces the output corresponding to target signatures to be prominent. This constraint ensures that the detected targets are accurately represented in the final output. It represents the 3-D target cube used as input to our detection model, where we enforce the model's output to be 1. This condition corresponds to the constraint placed on target outputs within the CEM framework, ensuring that detected targets are accurately represented. This aligns with the constraint on target output in the (7). By adapting the CEM framework, we have modified the loss function to better align with the requirements of our detection model. This adaptation ensures that the model not only excels in detecting targets accurately but also achieves robust background suppression, leading to improved detection performance and reliability. $\mathbf{O}(i, j)$ represents the distance between the spectrum at location (i, j) of the test image and the target prior spectrum, denoted as

$$\mathbf{O}(i, j) = \|\mathcal{X}(i, j) - \mathbf{d}\|_2^2 \tag{28}$$

where $||\cdot||_2$ represents the $L2$-norm. Consequently, a pixel's energy will decrease the closer it is to the target spectrum and vice versa. This effectively stops the target information from being lost.

In addition, the sparse loss constrains the abundance coefficients of each pixel to be sparse, meaning each pixel is composed of a sparse mixture of endmembers. Incorporating sparse constraints can improve the model's interpretability, concision, and consistency with reality. We use the $L1$-norm enforcing the sparsity of the abundance, denoted as

$$\mathcal{L}_{\text{sparse}} = \frac{1}{n} \sum_{i=1}^{n} ||\mathbf{S}(:, i)||_1 \tag{29}$$

where $\mathbf{S}(:, i)$ represents represents the abundance coefficients of the ith pixel and $||\cdot||_1$ represents $L1$-norm.

Combining the three aforementioned losses, we can obtain the total loss, represented as

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \alpha_2 \mathcal{L}_{\text{CEM}} + \alpha_3 \mathcal{L}_{\text{sparse}} \tag{30}$$

where α_2 and α_3 are the balance parameter.

IV. Experimental Evaluations

To validate the effectiveness of our proposed approach, experiments were conducted on six real HSI datasets. In this section, we first provide a detailed introduction to the hyperspectral datasets used in the experiments. Then, we present the detection results of our method. Subsequently, the performance of the proposed deep detection model is compared with several widely recognized advanced target detection methods, including CEM, ACE, CSCR, hCEM, DSC, HTD-net, background learning based on target suppression constraint (BLTSC), hyperspectral target detection based on interpretable representation network (HTD-IRN), triplet spectralwise transformer network for hyperspectral target detection (TSTDD), and information entropy estimation based on point-set topology (IEEPST). Finally, we evaluate the effectiveness of each loss and module through ablation experiments.

A. Experimental Configurations

1) Datasets: Five commonly used public datasets for evaluating HSI target detection methods are employed. To further enhance the generalization capabilities of the proposed method, a public dataset for evaluating HSI classification methods is also employed. Their details and included targets are summarized below.

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1) *San Diego [20]:* This dataset contains 224 spectral bands and covers wavelengths from 370 to 2510 nm. It covers 100 × 100 pixels. After removing noise bands, 186 bands were utilized. Three planes were the desired targets.
2) *Beach [46]:* This dateset was captured by the airborne visible infrared imaging spectrometer (AVIRIS) sensor. It covers 100 × 100 pixels with a spatial resolution of 4.4 m. After removing noise bands, 188 bands were utilized in our experiments. The target of interest in the image is a boat, occupying 11 pixels.
3) *Airport [46]:* The Airport dataset was collected by AVIRIS at the Gulfport Airport in Mississippi, USA. It covers 100 × 100 pixels with a spatial resolution of 3.4 m. After removing noise bands, 191 bands were utilized in our experiments. The targets of interest are three airplanes, totaling 88 pixels.
4) *Urban-I [46]:* This dataset originates from the AVIRIS sensor on an airborne platform. The image consists of 100 × 100 pixels with a spatial resolution of 17.2 m/pixel. After removing interference bands and noise bands, the final image consists of 207 bands. It focuses on 20 oil tanks of different sizes as the target of interest.
5) *Urban-II [46]:* This dataset was also collected by the AVIRIS sensor on the airborne platform. The image contains 100 × 100 pixels with a spatial resolution of 7.1 m. It also includes 205 spectral bands with a spectral resolution of 10 nm. There are 25 oil tanks of different sizes treated as targets.
6) *Indian Pines [47]:* This dataset consists of 145 × 145 pixels with a spatial resolution of 20 m. After removing 20 noise bands, the image consists of 200 bands. Here we select class 16, which contains fewer pixels, as the target, regarding the other pixels as the background.

2) Baselines: We compare with the following popular HSI detectors. CEM [12] minimizes the total energy output under the premise of ensuring a constant constraint on the target output. ACE [10] is based on the Gaussian distribution probability model. CSCR [19] employs sparse representations of target atoms and collaborative representations of background atoms to formulate test pixels. DSC [20] is based on clustering dictionary generation and sparse representation. HTD-net [33] is based on target sample generation and a convolutional neural network. BLTSC [36] is based on target suppression constraints and background learning. HTD-IRN [30] is a target detection method for spectral decomposition of linear mixture models. TSTDD [23] is a novel triplet spectral-wise transformer-based target detector for HTD tasks and uses data enhancement methods to build sufficient and balanced training samples. IEEPST [48] is a target detection method based on information entropy estimation.

All codes are downloaded from the authors' public homepages, and the hyperparameters are adjusted to the optimal according to the corresponding guidance in their articles. All datasets' gray values were normalized to the interval [0, 1] before the evaluations.

3) Evaluation Criteria: To assess the detection performance, we make use of the visible detection map and the receiver operating characteristic (ROC) curve [49] produced by varying a threshold τ. Target pixel numbers with detection values more than or less than τ are represented in 3-D ROC curves as true positive (TP) and false negative (FN), respectively. Similarly, false positive (FP) and true negative (TN) reflect the amounts of background pixels with detection values more than and less than τ, respectively. The effectiveness of target detection, background suppression, and overall detection of various methods is represented by the 2-D ROC curves $(\tau, \mathrm{P_d})$, $(\tau, \mathrm{P_f})$, and $(\mathrm{P_f}, \mathrm{P_d})$, respectively. Specifically, at each threshold, the false alarm probability $\mathrm{P_f}$ and the detection probability $\mathrm{P_d}$ are calculated by

$$\mathrm{P_d} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad \mathrm{P_f} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}}. \tag{31}$$

The ideal 2-D ROC curves $(\mathrm{P_f}, \mathrm{P_d})$, $(\tau, \mathrm{P_f})$, and $(\tau, \mathrm{P_d})$ should approach the upper left, lower left, and upper right corners of the coordinate axis, respectively. The $(\mathrm{P_f}, \mathrm{P_d})$ curve illustrates the relationship between the false alarm rate and detection rate. For quantitative evaluation, the areas under the 2-D ROC curves are computed and are represented as AUC $(\mathrm{P_f}, \mathrm{P_d})$, AUC $(\tau, \mathrm{P_d})$, and AUC $(\tau, \mathrm{P_f})$. In the case of AUC $(\mathrm{P_f}, \mathrm{P_d})$, and AUC $(\tau, \mathrm{P_d})$, higher values signify superior results, where 1 is the optimal value. Conversely, in the case of AUC $(\tau, \mathrm{P_f})$, a lower value is preferable, where 0 is the optimal value. Moreover, we take into account two composite criteria that integrate all three AUC values for a comprehensive assessment. These criteria are denoted as $\mathrm{AUC_{OA}}$ and $\mathrm{AUC_{SNPR}}$, and they are expressed as follows:

$$\mathrm{AUC_{OA}} = \mathrm{AUC_{(P_f,P_d)}} - \mathrm{AUC_{(\tau,P_f)}} + \mathrm{AUC_{(\tau,P_d)}} \tag{32}$$

$$\mathrm{AUC_{SNPR}} = \frac{\mathrm{AUC_{(\tau,P_d)}}}{\mathrm{AUC_{(\tau,P_f)}}}. \tag{33}$$

Among these two criteria, higher values of $\mathrm{AUC_{OA}}$ and $\mathrm{AUC_{SNPR}}$ indicate better detection performance. The optimal values for these criteria are 2 and $+\infty$, respectively.

B. Comparative Results on HSI Target Detection

Fig. 5 illustrates the detection results of different methods on six real hyperspectral datasets. It can be observed that, compared to other methods, our method detects the target more prominently and achieves more effective background suppression. The detection results clearly show a significant separation between targets and background. This improvement is attributed to the capability of our BMM deep network to effectively mitigate the interference caused by nonlinear scattering, as well as the lightweight BSM we proposed. As a result, it can recover more complete and accurate target information, enhancing the overall performance of target detection. Classical detection methods like CEM and ACE struggle to maintain the integrity of target pixels while sufficiently suppressing the background. In addition, although IEEPST and HTD-net exhibit good integrity in target detection, their performance in background suppression is inadequate due to the complex background information in HSIs, resulting

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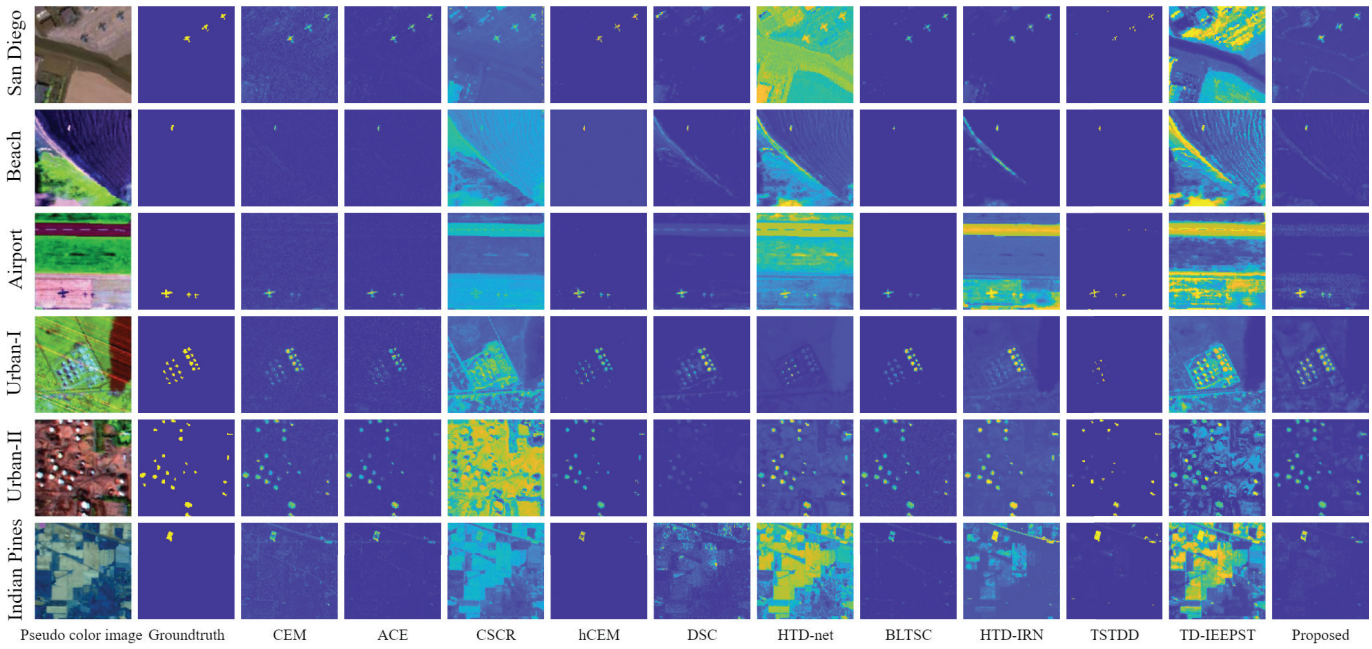


Fig. 5. Pseudo color images of the six datasets, along with their corresponding groundtruth images and the detection results from different methods.

in numerous false alarms. BLTSC, hCEM, and TSTDD successfully address the problem of excessive background response, resulting in smoother backgrounds. However, they tend to misclassify weak targets as unnecessary information. In comparison, our method demonstrates significantly greater differences between targets and background in the results, especially on the Airport and Urban-II datasets.

In addition to the visualized detection results, Fig. 6 illustrates the ROC curves of various methods on different datasets, providing a more precise assessment of the detection performance. From the results, it can be observed that our method consistently detects complete target pixels with low false alarm rates. Regarding overall detection performance, compared to other methods, our proposed approach exhibits ROC curves (P_f, P_d) closer to the top left corner of the Beach, Airport, Urban-I, and Urban-II datasets. However, when P_f is less than 10^{-3}, our method shows slightly lower accuracy in this quantitative metric compared to hCEM on the San Diego dataset and TSTDD on the Indian Pines dataset. Nevertheless, the detection results demonstrate that our method achieves a clearer separation between targets and background. Furthermore, our method outperforms these methods in terms of the target detection completeness. Overall, although our method is not the most outstanding in detecting target completeness, it still outperforms most comparative methods and achieves a better balance between detecting target completeness and background suppression. In summary, our proposed method consistently achieves the highest average detection rates in all six datasets.

To further quantitatively analyze the detection results, this study employs five AUC metrics to evaluate the detection performance of each method across the six datasets. Table I presents the AUC results of different methods. These outcomes allow for further quantitative analysis of the detection results. Clearly, our method attains the highest $AUC_{(P_f,P_d)}$ across all six HSI datasets, demonstrating its competitiveness. This improvement is attributed to our proposed U-PSA module, which effectively captures and extracts both the spectral and spatial information from the image, leading to more comprehensive and accurate feature representation. Concerning the AUC_{OA} metric describing overall detection performance, our method achieves the best results on the San Diego, Beach, and Airport datasets, and gets the suboptimal result on the Urban-I dataset. Owing to the capability of our BMM deep model to effectively avoid interference from nonlinear scattering during the imaging process, our approach can recover more realistic and complete imaging targets while ensuring sufficient background suppression. This ensures higher fidelity in target representation and enhances the overall accuracy of detection. In addition, concerning the $AUC_{(\tau,P_d)}$ metric describing target detection effectiveness, our method attains the suboptimal result on the Urban-I dataset. Furthermore, although our method does not achieve the best results in the $AUC_{(\tau,P_f)}$ metric, it surpasses most comparative methods. Nevertheless, the experimental results demonstrate that, compared with other methods, our method achieves the best balance between target detection integrity and background suppression. Therefore, our method significantly outperforms other approaches regarding the overall detection level.

According to the detection result, while our approach excels in detecting complete targets and separating targets from backgrounds, it does not achieve the optimal false alarm rates. The CEM loss function plays a critical role in our model but has inherent limitations. Increasing the weight of the target constraint term enhances target detection but weakens background suppression, leading to higher false alarms. Conversely, emphasizing the output energy minimization term improves background suppression and reduces false alarm rates, but this diminishes the effectiveness of target detection.

To address the issue of increased false alarm rates, we have additionally incorporated the previously mentioned BSM,

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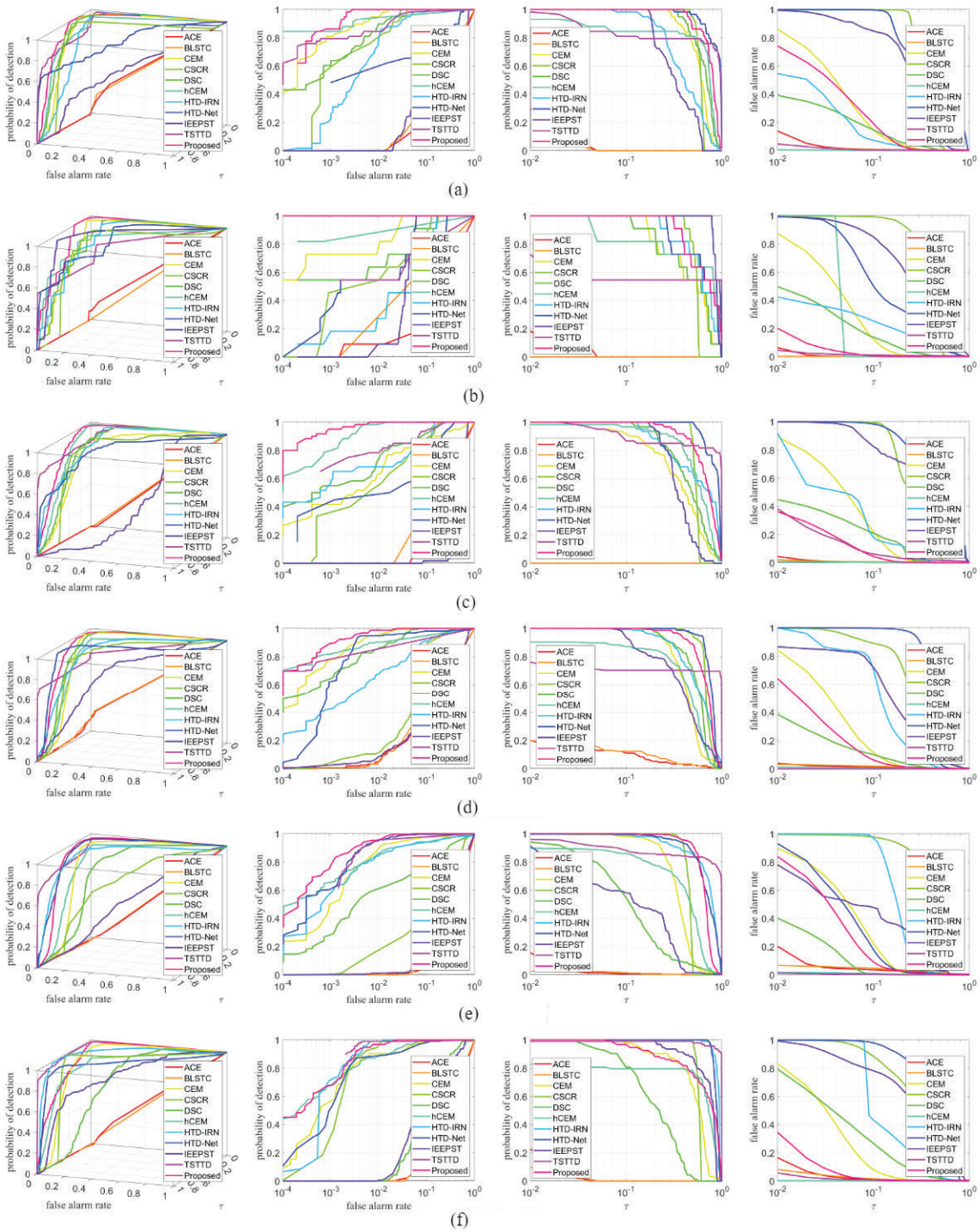


Fig. 6. 3-D ROC curves and the corresponding 2-D ROC curves for various comparative methods on six HSIs. (Left to right) 3-D ROC curve, 2-D ROC (P_f, P_d) curve, 2-D ROC(τ, P_d) curve, and 2-D ROC(τ, P_f) curve, respectively. (a) San Diego. (b) Beach. (c) Airport. (d) Urban-I. (e) Urban-II. (f) Indian Pines.

which aims to balance background suppression and the completeness of target detection. By integrating the BSM, our method strikes a better balance between target detection completeness and background suppression. This improvement not only enhances the overall performance of detection but also ensures the reliability and practicality of the detection results. We believe that this strategy is reasonable and effective, providing a more comprehensive and accurate target detection outcome suitable for various applications.

In Table I, we include detection time as an additional performance metric in our comparative analysis. While our method may exhibit slightly longer detection time compared to some lightweight models, it offers significant advantages in target completeness and robustness. The increased computational

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TABLE I

QUANTITATIVE PERFORMANCE RESULTS OBTAINED BY ALL METHODS ON SIX DATASETS. OPTIMAL AND SUBOPTIMAL ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Dataset		CEM	ACE	CSCR	hCEM	DSC	IEEPST	HTD-Net	BLSTC	HTD-IRN	TSTDD	Proposed
San Diego	$AUC_{(P_f,P_d)}$	0.9939	0.9932	0.9919	0.9651	0.9929	0.9235	0.9775	0.9957	0.9981	0.9949	**0.9997**
	$AUC_{(\tau,P_d)}$	0.6003	0.5402	0.5689	0.6494	0.2265	**0.8857**	0.8551	0.4588	0.4579	0.4236	0.7749
	$AUC_{(\tau,P_f)}$	0.0415	0.0051	0.1586	0.0017	0.0076	0.3714	0.6166	0.0010	0.0062	**0.0005**	0.0560
	AUC_{OA}	1.5527	1.5283	1.4022	1.6128	1.2118	1.4378	1.2160	1.4535	1.4498	1.4179	**1.7186**
	AUC_{SNPR}	14.4651	105.9216	3.5870	382.0000	29.8026	2.3849	1.3868	458.8000	73.8548	**795.7165**	13.8375
	Time	**0.015**	0.026	6.060	0.027	4.033	0.281	18.115	0.112	2.786	4.198	2.922
	Params/M	-	-	-	-	-	-	**0.288**	19.978	1.582	3.471	6.626
	GFLOPs	-	-	-	-	-	-	**3.181**	7.076	4.404	9.730	6.412
Beach	$AUC_{(P_f,P_d)}$	0.9523	0.9919	0.9607	0.9544	0.9864	0.9765	0.9200	**1.0000**	0.9926	0.9674	**1.0000**
	$AUC_{(\tau,P_d)}$	0.4376	0.5129	0.5446	0.6915	0.5501	**0.9158**	0.7459	0.3732	0.6508	0.5454	0.7298
	$AUC_{(\tau,P_f)}$	0.0080	0.0027	0.3590	0.0412	0.0125	0.3037	0.1920	**0.0000**	0.0152	0.0002	0.0084
	AUC_{OA}	1.3819	1.5021	1.1463	1.6047	1.5240	1.5886	1.4739	1.3732	1.6282	1.5127	**1.7214**
	AUC_{SNPR}	54.7000	189.9630	1.5170	16.7840	44.0080	3.0153	3.8849	$+\infty$	42.8158	3636.267	86.8810
	Time	**0.016**	0.025	6.112	0.187	54.527	2.744	13.567	0.107	2.617	4.227	2.950
	Params/M	-	-	-	-	-	-	**0.288**	19.903	1.581	3.459	6.519
	GFLOPs	-	-	-	-	-	-	**3.146**	7.061	4.401	9.665	6.387
Airport	$AUC_{(P_f,P_d)}$	0.9839	0.9770	0.9878	0.9915	0.9912	0.9699	0.7681	0.9841	0.9803	0.9959	**0.9997**
	$AUC_{(\tau,P_d)}$	0.4299	0.3490	0.5589	0.5924	0.4308	**0.9291**	0.6479	0.3345	0.9019	0.5424	0.7083
	$AUC_{(\tau,P_f)}$	0.0198	0.0052	0.2905	0.0044	0.0140	0.4347	0.4710	**0.0006**	0.3636	0.0009	0.0212
	AUC_{OA}	1.3940	1.3208	1.2571	1.5798	1.4080	1.4642	0.9450	1.3180	1.5186	1.5375	**1.6868**
	AUC_{SNPR}	21.7121	67.1154	1.9270	134.7045	30.7714	2.1371	1.3756	557.5000	2.4805	**596.1099**	33.4104
	Time	**0.018**	0.028	6.079	0.293	13.956	0.724	18.922	0.133	2.845	4.176	2.972
	Params/M	-	-	-	-	-	-	**0.288**	19.903	1.583	3.495	6.463
	GFLOPs	-	-	-	-	-	-	**3.201**	7.108	4.412	9.858	6.382
Urban-I	$AUC_{(P_f,P_d)}$	0.9888	0.9839	0.9546	0.9538	0.9976	0.9975	0.9142	0.9924	0.9896	0.9981	**0.9993**
	$AUC_{(\tau,P_d)}$	0.4919	0.3399	0.6319	0.4511	0.4744	**0.8729**	0.1741	0.4791	0.4977	0.1201	0.6944
	$AUC_{(\tau,P_f)}$	0.0172	0.0107	0.2779	0.0057	0.0075	0.1978	0.0350	0.0019	0.0475	**0.0000**	0.0328
	AUC_{OA}	1.4635	1.3131	1.3086	1.3992	1.4645	**1.6727**	1.0533	1.4696	1.4398	1.1183	1.6609
	AUC_{SNPR}	28.5988	31.7664	2.2738	79.1404	63.2533	4.4135	4.9743	252.1579	10.4779	**5668.7900**	21.1707
	Time	**0.024**	0.032	6.162	0.249	191.184	6.291	18.878	0.121	3.072	4.896	2.917
	Params/M	-	-	-	-	-	-	**0.288**	20.908	1.594	3.698	6.424
	GFLOPs	-	-	-	-	-	-	**3.475**	7.364	4.454	10.988	6.402
Urban-II	$AUC_{(P_f,P_d)}$	0.9710	0.9951	0.7648	0.9463	0.9349	0.9900	0.9955	0.9937	0.9699	0.9966	**0.9986**
	$AUC_{(\tau,P_d)}$	0.4907	0.4235	**0.7492**	0.4194	0.0740	0.6351	0.7053	0.5472	0.6552	0.4163	0.6639
	$AUC_{(\tau,P_f)}$	0.0256	0.0078	0.6297	0.0027	0.0034	0.1277	0.0732	0.0245	0.0884	**0.0018**	0.0503
	AUC_{OA}	1.4361	1.4108	0.8843	1.3630	1.0055	1.4974	**1.6276**	1.5164	1.5367	1.4110	1.6122
	AUC_{SNPR}	19.1680	54.2949	1.1898	155.3333	21.7647	4.9731	9.6352	22.3347	7.4188	**222.2270**	31.2604
	Time	**0.017**	0.035	6.153	0.235	192.979	3.063	21.236	0.115	3.217	4.639	13.199
	Params/M	-	-	-	-	-	-	**0.288**	20.704	1.592	3.672	6.587
	GFLOPs	-	-	-	-	-	-	**3.454**	7.332	4.448	10.775	6.446
Indian Pines	$AUC_{(P_f,P_d)}$	0.9939	0.9963	0.9977	0.9137	0.8461	0.9980	0.9919	0.9939	0.9987	0.9983	**0.9991**
	$AUC_{(\tau,P_d)}$	0.5386	0.4964	0.5734	0.6177	0.2296	**0.9729**	0.8705	0.4504	0.9157	0.9702	0.7292
	$AUC_{(\tau,P_f)}$	0.0447	0.0073	0.2967	**0.0044**	0.0838	0.3772	0.4696	0.0068	0.1615	0.0073	0.0147
	AUC_{OA}	1.4878	1.4855	1.2743	1.5270	0.9919	1.5937	1.3928	1.4374	1.7529	**1.9612**	1.7136
	AUC_{SNPR}	12.0492	67.8589	1.9324	**140.3864**	2.7399	2.5790	1.8537	66.1949	5.6700	132.7168	37.6158
	Time	**0.063**	0.122	8.772	0.557	141.5268	7.941	26.979	0.266	4.926	6.693	3.341
	Params/M	-	-	-	-	-	-	**0.288**	20.013	1.589	3.608	6.609
	GFLOPs	-	-	-	-	-	-	**3.349**	8.515	9.261	20.889	13.457

demand is primarily due to the detailed modeling of nonlinear scattering by BMM.

However, compared to the HTD-IRN method based on an LMM, our method demonstrates higher efficiency on the Urban-I, Urban-II, and Indian Pines datasets. This indicates that our U-PSA module is superior in detection efficiency. In addition, when compared to other deep learning-based methods such as HTD-net and TSTDD, our method exhibits significantly shorter detection times. Furthermore, to more accurately compare the efficiency of our method with that of other methods, we introduced the evaluation metrics of parameter count and floating point operations (FLOPs). HTD-Net consistently exhibits the smallest model size, with a fixed parameter count of 0.288M on all six datasets. This reflects an extremely lightweight architecture. However, such compactness is achieved at the expense of detection accuracy, as HTD-Net delivers relatively lower $AUC_{(P_f,P_d)}$ in comparison to other methods. In contrast, BLSTC relies on a large model, with parameter counts reaching up to 20.908M on the Urban-I dataset. This increase in network capacity is accompanied by a substantial rise in computational demands. TSTDD incurs GFLOPs exceeding ten on Urban-I, Urban-II, and Indian Pines datasets. High computational overhead can limit their applicability in resource-constrained or real-time systems. Our method effectively balances model compactness and computational efficiency. While its parameter counts range

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TABLE II

AUC SCORES FOR DIFFERENT LOSS FUNCTIONS USED ON SIX DATASETS. OPTIMAL AND SUBOPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Dataset		L_r	L_{r+s}	L_{r+C}	L_{r+s+C}
San Diego	$AUC_{(P_f,P_d)}$	0.9096	0.9189	0.9967	**0.9995**
	$AUC_{(\tau,P_d)}$	0.4420	0.3652	0.5290	**0.7749**
	$AUC_{(\tau,P_f)}$	0.1067	0.0677	**0.0178**	0.0560
Beach	$AUC_{(P_f,P_d)}$	0.8033	0.8908	0.9849	**1.0000**
	$AUC_{(\tau,P_d)}$	0.5662	0.5184	0.2422	**0.7298**
	$AUC_{(\tau,P_f)}$	0.0231	0.1469	**0.0829**	0.0212
Airport	$AUC_{(P_f,P_d)}$	0.8555	0.9867	0.9648	**0.9995**
	$AUC_{(\tau,P_d)}$	0.4271	0.6786	0.3501	**0.7083**
	$AUC_{(\tau,P_f)}$	0.0231	0.1469	0.0829	**0.0212**
Urban-I	$AUC_{(P_f,P_d)}$	0.6472	0.9617	0.9969	**0.9982**
	$AUC_{(\tau,P_d)}$	0.4224	0.5436	0.6276	**0.6944**
	$AUC_{(\tau,P_f)}$	0.3059	0.0719	0.0333	**0.0328**
Urban-II	$AUC_{(P_f,P_d)}$	0.8667	0.8863	0.9983	**0.9986**
	$AUC_{(\tau,P_d)}$	0.5202	0.5016	0.4938	**0.6639**
	$AUC_{(\tau,P_f)}$	0.6041	0.3101	0.0624	**0.0503**
Indian Pines	$AUC_{(P_f,P_d)}$	0.8164	0.9288	0.9880	**0.9991**
	$AUC_{(\tau,P_d)}$	0.5176	0.5635	0.4906	**0.7292**
	$AUC_{(\tau,P_f)}$	0.2792	0.1686	0.0287	**0.0147**

from 6.424M to 6.626M, which are substantially lower than BLSTC, our method obtains the highest $AUC_{(P_f,P_d)}$ on all six datasets. On the Beach dataset, our method achieves a $AUC_{(P_f,P_d)}$ of 1.0000 with only 6.387 GFLOPs, compared to TSTDD, which achieves 9.665 GFLOPs. Compared with the LMM-based model HTD-IRN, the increase in the number of parameters in our method is primarily attributed to the virtual endmember construction, which enhances spectral representation capabilities. The increased GFLOPs mainly result from the BMM modeling. We consider these additional computational costs to be acceptable given the consistent improvements in detection accuracy on all six datasets.

In complex applications with a high number of second-order interactions, the performance of BMM is notably superior to other deep learning-based methods. This also confirms the unique advantages of BMM in the field of nonlinear modeling. Specifically, BMM has the ability to effectively capture complex second-order ground object interactions and nonlinear scattering, thus enhancing the performance of target detection. Although virtual endmembers of BMM lead to an increase in the number of parameters and computational complexity, the interactions of the third-order and above can be ignored; this increased cost can be fully utilized to achieve the optimal balance between computational consumption and performance. Overall, our method consistently produces satisfactory visual detection maps, achieves outstanding quantitative results, and demonstrates promising target-background separation capabilities across all six datasets.

C. Ablation Study and Parameter Analysis

1) Effectiveness Evaluation of Different Loss Terms: To verify the effectiveness of each loss function term, we conducted ablation experiments, and the results are shown in the Table II. Here, $L_r = L_{\text{recon}}$, $L_{r+s} = L_{\text{recon}} + L_{\text{sparse}}$, $L_{r+C} = L_{\text{recon}} + L_{\text{CEM}}$, and $L_{r+s+C} = L_{\text{recon}} + L_{\text{sparse}} + L_{\text{CEM}}$. It is evident that the CEM loss plays a critical role in the detection model. When only reconstruction loss is considered without the CEM loss, our proposed method fails to produce convincing outcomes. However, when CEM loss is combined with reconstruction loss, there is a significant improvement in overall detection performance as measured by $AUC_{(P_f,P_d)}$. This strongly demonstrates the effectiveness of the CEM loss. Furthermore, while using only CEM and reconstruction losses significantly improves performance, there remains a gap compared to some advanced detectors. Therefore, we introduced sparsity loss. The purpose of sparse representation is to use the minimum number of most relevant bases to represent each pixel, which aids in distinguishing between targets and backgrounds. Consequently, sparsity loss can further enhance the performance of deep detectors. Combining all three losses yields better detection results compared to their individual impacts.

TABLE III

QUANTITATIVE DETECTION RESULTS OF LMM AND BMM BY AUC ON SIX DATASETS. OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD

Dataset		$AUC_{(P_f,P_d)}$	$AUC_{(\tau,P_d)}$	$AUC_{(\tau,P_f)}$
San Diego	LMM	0.9811	0.2868	**0.0164**
	BMM	**0.9997**	**0.7749**	0.0560
Beach	LMM	0.9979	0.5037	0.0152
	BMM	**1.0000**	**0.7298**	**0.0084**
Airport	LMM	0.9978	**0.8011**	0.1194
	BMM	**0.9997**	0.7083	**0.0212**
Urban-I	LMM	0.9916	**0.8021**	0.2414
	BMM	**0.9993**	0.6944	**0.0328**
Urban-II	LMM	0.9947	0.4938	0.0624
	BMM	**0.9986**	**0.6639**	**0.0503**
Indian Pines	LMM	0.9925	0.6196	0.0786
	BMM	**0.9991**	**0.7292**	**0.0147**

Notably, for $AUC_{(\tau,P_f)}$, the combination of all three losses achieves the best result, underscoring their collective promotion of target detection. In particular, regarding background suppression as indicated by $AUC_{(\tau,P_f)}$, it is clear that adding the CEM loss leads to a substantial decrease in scores. This indicates that the CEM loss has an extremely effective role in suppressing background noise. The ablation study confirms that the three proposed loss terms not only promote the detection of salient targets but also significantly enhance the network's ability to suppress backgrounds. In summary, the proposed loss functions are effective, and each component is indispensable to achieving optimal detection performance.

2) Effectiveness Evaluation of BMM: To validate the effectiveness of the BMM, this study compared it with the LMM through experimental comparisons. Table III presents the detection results of different datasets using the two theoretical models for spectral decomposition. The first row in the table shows the results of experiments based on LMM, and the second row shows the results of experiments based on our proposed approach based on BMM. From the table, it is evident that on the Beach and Urban-II datasets, the detection model based on the BMM outperforms the LMM-based detection model across all AUC metrics.

Furthermore, on the San Diego dataset, although our method slightly lags behind the LMM-based model in terms of $AUC_{(\tau,P_f)}$, it achieves more convincing results on other AUC scores. For the Airport and Urban-I datasets, except for the $AUC_{(\tau,P_d)}$ score describing target detection integrity, the

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TABLE IV
AUC SCORES FOR DIFFERENT NETWORK MODULES USED ON SIX DATASETS. OPTIMAL AND SUBOPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Dataset		SEblock	ViT	CBAM	UNet	UPSA
San Diego	$AUC_{(P_f,P_d)}$	0.9867	0.9875	0.9953	0.9942	**0.9997**
	$AUC_{(\tau,P_d)}$	0.5066	0.5289	0.6623	0.5392	**0.7749**
	$AUC_{(\tau,P_f)}$	**0.0302**	0.0541	0.0814	0.0311	0.0560
Beach	$AUC_{(P_f,P_d)}$	0.9879	0.9973	0.9181	0.9883	**1.0000**
	$AUC_{(\tau,P_d)}$	0.5117	0.7064	0.4518	0.5472	**0.7298**
	$AUC_{(\tau,P_f)}$	0.0171	0.0304	0.0747	0.0162	**0.0084**
Airport	$AUC_{(P_f,P_d)}$	0.9747	0.9656	0.9952	0.9739	**0.9997**
	$AUC_{(\tau,P_d)}$	**0.7731**	0.7511	0.6931	0.7724	0.7083
	$AUC_{(\tau,P_f)}$	0.1767	0.1557	0.0396	0.1515	**0.0212**
Urban-I	$AUC_{(P_f,P_d)}$	0.9969	0.9968	0.9955	0.9926	**0.9993**
	$AUC_{(\tau,P_d)}$	0.6471	0.6706	0.6467	0.6583	**0.6944**
	$AUC_{(\tau,P_f)}$	0.0807	0.0433	0.0474	0.0689	**0.0328**
Urban-II	$AUC_{(P_f,P_d)}$	0.9957	0.9955	0.9948	0.9624	**0.9986**
	$AUC_{(\tau,P_d)}$	0.4878	0.5276	0.0927	0.3919	**0.6639**
	$AUC_{(\tau,P_f)}$	0.1140	0.3908	0.2080	**0.0259**	0.0503
Indian Pines	$AUC_{(P_f,P_d)}$	0.9883	0.9884	0.9797	0.9822	**0.9991**
	$AUC_{(\tau,P_d)}$	0.6273	0.6791	0.5514	0.6239	**0.7292**
	$AUC_{(\tau,P_f)}$	0.0714	0.1225	0.0779	0.0464	**0.0147**

detection results of the proposed method are better than the detection model based on LMM. While the LMM-based method may ensure better target integrity on Airport and Urban-I datasets, their $AUC_{(\tau,P_f)}$ scores are higher than our method, especially on the Urban-I dataset, which exceeds 0.2, indicating unacceptable errors in background suppression.

Through comparative experiments, we verified the effectiveness of the BMM in enhancing detection performance within our detection framework. HSIs often exhibit a wide range of nonlinear features, limiting the detection capabilities of conventional LMM-based detectors. The BMM enhances the detector's ability to recognize nonlinear features by simulating interactions of various materials exhibiting nonlinear scattering through the addition of a virtual endmember matrix. Therefore, our deep network detection model based on BMM successfully separates targets from backgrounds in complex HSIs.

3) Effectiveness Evaluation of Feature Extraction Module: To validate the effectiveness of our proposed multiscale feature extraction network, U-PSA, we conducted performance comparisons by substituting U-PSA with four existing advanced network modules. Table IV presents the AUC scores comparison results between our multiscale feature extraction module and other state-of-the-art network modules. The compared network modules include SE-block [50], convolutional block attention module (CBAM) [51], vision transformer (ViT) [52], and U-net [53]. These modules have been proven effective across a range of computer vision tasks. SE-block adaptively learns channel-wise attention weights to focus on relevant features. ViT captures nonlocal spatial features through multihead self-attention mechanisms. CBAM combines spatial and channel attention to extract features effectively. U-net employs an encoder–decoder structure to extract multiscale features.

Among these advanced modules, spatial attention-based modules can fully utilize spatial information between neighboring pixels, whereas channel attention-based modules excel at adaptively learning discriminative features from different spectral channels. Our proposed multiscale feature extraction module U-PSA, integrates both spatial and channel attention, enabling the network to not only effectively extract spectral features from complex HSIs but also leverage spatial information within the images.

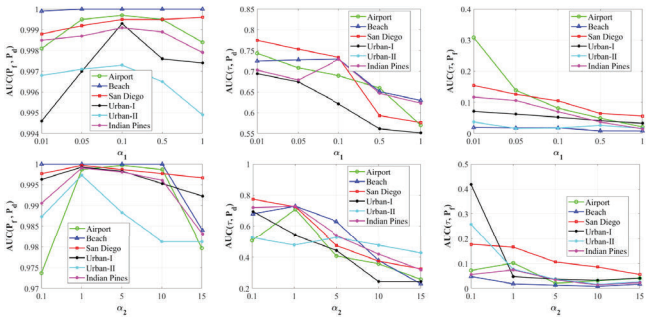


Fig. 7. $AUC_{(P_d,P_f)}$, $AUC_{(\tau,P_d)}$, and $AUC_{(\tau,P_f)}$ for different datasets across various settings of the parameters α_1 and α_2.

From Table IV, it is evident that our proposed multiscale feature extraction module outperforms other advanced network modules in terms of $AUC_{(P_f,P_d)}$ scores across six datasets. This validates the superior capability of U-PSA in extracting more accurate abundance and enhancing detection performance. In addition, our U-PSA module achieves the best $AUC_{(\tau,P_d)}$ results on five datasets, indicating its superior performance in target detection. Furthermore, in the background suppression metric $AUC_{(\tau,P_f)}$, our model obtains the best scores on five datasets, underscoring its effectiveness in suppressing background noise.

In summary, the proposed U-PSA module is particularly well-suited for processing HSIs. Its ability to integrate spectral and spatial feature extraction provides a comprehensive approach to handling high-dimensional data, leading to enhanced overall performance.

4) Parameter Selection: The parameter α_1 determines the weight of the energy minimization component in the CEM loss function. A larger α_1 enhances the model's ability to suppress background, while a smaller α_1 increases the prominence of target detection. As illustrated in Fig. 7, increasing α_1 strengthens background suppression but can lead to partial suppression of targets. Conversely, reducing α_1 highlights targets more effectively but increases the false alarm rate. The $AUC_{(\tau,P_d)}$ and $AUC_{(\tau,P_f)}$ provide insights into the trade-offs between background suppression and target highlighting. When α_1 is within the range of 0.1–1, $AUC_{(\tau,P_f)}$ decreases gradually, while $AUC_{(\tau,P_d)}$ drops significantly. This indicates that we can sacrifice a small amount of background suppression to maximize target highlighting. Based on these observations, we set $\alpha_1 = 0.1$ to achieve a balance that emphasizes target detection with minimal impact on background suppression.

The parameter α_2 controls the overall weight of the CEM loss function, influencing the separation between target and background components. Fig. 7 shows the effect of different α_2 values on detection performance across multiple datasets. When $\alpha_2 = 1$, the $AUC_{(P_d,P_f)}$ is notably higher compared to other values, indicating optimal target-background separation. Increasing α_2 beyond 1 leads to a decrease in both $AUC_{(\tau,P_d)}$ and $AUC_{(\tau,P_f)}$, as the output energy tends to become smaller, reducing overall detection performance. Specifically, when

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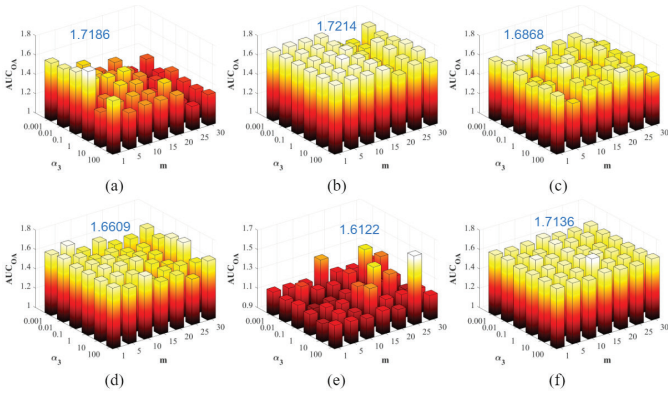


Fig. 8. Effect of parameters m and α_3 for our methods. (a) San Diego. (b) Beach. (c) Airport. (d) Urban-I. (e) Urban-II. (f) Indian Pines.

$\alpha_2 > 1$, $\mathrm{AUC}_{(\tau,\mathrm{P_d})}$ decreases significantly, while $\mathrm{AUC}_{(\tau,\mathrm{P_f})}$ remains relatively stable. Given these findings, we set $\alpha_2 = 1$ to achieve the best balance between target-background separation and overall detection accuracy.

Given that the number of background endmembers and the sparsity of targets vary across different datasets, it is necessary to tailor the settings of parameters m and α_3 specifically for each dataset. To achieve the best balance between the integrity of target detection and the sufficiency of background suppression, we chose the $\mathrm{AUC_{OA}}$, which is more intuitive and comprehensive, as the evaluation metric. The results are presented in the bar chart shown in Fig. 8. It can be observed that variations in m and α_3 significantly influence the optimal detection performance, with the optimal combination yielding the highest $\mathrm{AUC_{OA}}$ as indicated by the blue markers. The optimal values of parameters m and α_3 are contingent upon the intrinsic properties of the dataset. For the San Diego dataset, the highest $\mathrm{AUC_{OA}}$ is achieved when m is 1. This indicates that the background in the San Diego dataset is not complex, and a single spectral basis is sufficient for accurate representation due to its simple and homogeneous backgrounds. Conversely, for the Urban-II and Indian Pines datasets, the best detection performance is attained when their m are 25 and 15, respectively. This suggests that these datasets possess more complex backgrounds, necessitating a greater number of diverse spectral bases to adequately construct the background subspace. In addition, increasing m beyond this point results in a slight decline in detection accuracy, as an excess of background bases can interfere with target detection. Moreover, the sparsity component is closely related to the diversity of the spectral bases. This relationship underscores the importance of balancing the number of spectral bases used to ensure optimal detection performance while maintaining the necessary level of sparsity. In summary, the choice of m should be carefully considered based on the complexity of the dataset's background, with simpler datasets benefiting from fewer spectral bases and more complex datasets requiring a larger number, but not excessively so, to preserve detection accuracy. In addition, when the value of m is small, the sparse coefficient α_3 should take a smaller value; when m is large, the dataset has higher requirements for sparsity, and the value of α_3 should be larger. Based on the above experiments and analysis of the experimental results, we, respectively, set (m, α_3) as $(1, 1)$, $(10, 1)$, $(10, 0.01)$, $(5, 0.001)$, $(25, 100)$, and $(15, 10)$ for six datasets, respectively.

V. Conclusion and Future Work

This study proposes a spectral decomposition deep network, which intuitively implements the BMM detection framework. Specifically, the BMM is transformed into a deep neural network to decompose the image into coefficient subspaces of targets and backgrounds. To further extract and combine the spectral and spatial information of the image, and more accurately learn the distribution of targets and backgrounds, a multiscale feature extraction module U-PSA, is proposed to capture the nonlinear mapping between HSI and the abundance coefficient. In addition, CEM loss and sparse loss are introduced to make the separation of background and targets more accurate. Experimental comparisons on six real HSI datasets validate the method's good balance between highlighting targets and suppressing background. Moreover, quantitative and qualitative results such as visual detection maps, ROC curves, and AUC scores indicate that the proposed method outperforms other competing methods. The effectiveness of the loss terms, BMM, and U-PSA module is verified through ablation experiments. In future work, our goal is to incorporate richer prior knowledge to guide the deep network in expressing HSIs better. For example, we plan to consider incorporating low-rank properties as an additional term in the network to further enhance the model's functionality and capacity for generalization. These efforts will help advance the field of HSI processing and provide better solutions for tasks such as target detection and background separation.

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